

Sudha Madhuri Vinnakota

AI Engineer

Hyderabad, India | +916300962337 | sudhamadhurivinnakota@gmail.com
LinkedIn: www.linkedin.com/in/sudha-ai | GitHub: github.com/sudha-madhuribolloju
Portfolio: sudha-madhuribolloju.github.io/sudha-ai-portfolio/

PROFESSIONAL SUMMARY

AI Engineer and problem-solving technologist with hands-on experience in Generative AI, LLMs, AI Agents, RAG, and intelligent automation. Skilled in Python, LangChain, LangGraph, CrewAI, MCP, AWS, and CI/CD. Focused on solving real-world challenges by building practical, reliable, and scalable AI solutions using APIs, automation, and workflow orchestration, with strengths in root-cause analysis and system design.

TECHNICAL SKILLS

Programming: Python

Generative AI: LLMs, Generative AI, OpenAI, Claude, Gemini, Prompt Engineering, RAG, AI Agents, Agentic AI

AI Frameworks: LangChain, LangGraph, CrewAI, FastAPI, Flask

AI & Integration: MCP, Tool Calling, API Integration, n8n, Webhooks

AI Development Tools: Cursor, GitHub Copilot, Antigravity, Google AI Studio, Firebase Studio, Lovable

Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC, RDS, CloudWatch), AWS Management Console

DevOps & CI/CD: Docker, Docker Hub, Git, GitHub, GitHub Actions, CI/CD, Kubernetes

Databases & Vector Stores: PostgreSQL, pgvector, Supabase

Web & APIs: REST APIs, JSON, HTTP/HTTPS

Productivity & Data Tools: Excel, Jupyter Notebook, Postman

WORK EXPERIENCE

AI Engineer | AgentOpsLabs

April 2026 – Present

- Design and develop LLM-powered applications, AI agents, and intelligent automation solutions integrating AI models with enterprise data, APIs, tools, and business workflows.
- Build AI applications using Python, LangChain, LangGraph, and CrewAI to transform manual business processes into intelligent, automated workflows.
- Design AI and multi-agent systems using CrewAI and LangGraph, enabling task reasoning, agent collaboration, state management, and tool execution.
- Develop MCP servers and tool integrations connecting AI agents with APIs, databases, external services, and enterprise systems.

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- Implement RAG-based solutions using semantic retrieval and external knowledge sources to deliver relevant, context-aware responses.
- Orchestrate end-to-end AI workflows combining LLM reasoning, retrieval, memory, tool calling, and workflow control.
- Develop Python-based AI backend services and REST APIs, integrating LLM applications with databases, third-party services, and enterprise platforms.
- Deploy and productionize AI applications on AWS, leveraging EC2, S3, Lambda, GitHub, and CI/CD pipelines for scalable and reliable deployments.
- Automate business workflows using n8n and no-code/low-code platforms, integrating multiple systems through event-driven workflows.
- Use AI-assisted development tools such as Antigravity to accelerate application development, debugging, experimentation, and delivery.
- Optimize agent orchestration, API interactions, retrieval workflows, and cloud resource utilization to improve AI application performance.
- Collaborate with cross-functional teams to understand business requirements and translate them into practical, deployable AI solutions.

UiPath Developer | Cybermatics Systems

June 2021 – December 2024

- Developed and maintained 10 UiPath automation bots across multiple projects using the REFramework.
- Automated business processes across web, Windows, mainframe, Excel, Word, and PDF-based applications.
- Designed and developed UiPath workflows and reusable objects based on client business requirements.
- Worked directly with clients for requirements gathering, process analysis, and solution design.
- Prepared automation documentation including PDD (Process Design Document) and SDD (Solution Design Document).
- Managed deployed bots through UiPath Orchestrator, including queues, assets, scheduling, monitoring, and maintenance.
- Supported end-to-end automation delivery from requirement gathering through development, deployment, monitoring, and maintenance.

PROJECTS:

1. Nexora AI Classroom | AI-Powered Learning Platform: AI-powered learning platform combining LLMs, RAG, NLP, speech recognition, and learning analytics for personalized learning.

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- Developed AI Tutor, RAG-based PDF Chat, lecture transcription, summarization, and automated quiz generation.
- Designed the end-to-end workflow: Lecture → Speech-to-Text → NLP → LLM → RAG → Quiz → Analytics.
- Built backend services using FastAPI, PostgreSQL/pgvector, with WebSockets, JWT authentication, and role-based access control.

Tech Stack: Python, FastAPI, PostgreSQL, pgvector, LLMs, RAG, NLP, WebSockets, JWT

2.AI Linux Automation Agent | Gemini + LangChain: Built an AI-powered Linux automation agent using Google Gemini and LangChain that converts natural-language requests into Linux commands and executes them through a Shell Tool. Implemented system information retrieval and deployed the agent on AWS EC2 Linux.

Tech Stack: Python, Gemini, LangChain, Linux, Shell, AWS EC2

3.Multi-Agent Technology Research & Observability System | CrewAI + Langfuse: Built a CrewAI multi-agent research workflow using specialized Researcher and Reporting Agents. Implemented Pydantic for structured and validated JSON outputs and integrated Langfuse for LLM observability, including latency, token usage, and agent context tracking.

Tech Stack: Python, CrewAI, Pydantic, Langfuse, LLMs

4.Smart Q&A Bot | LangChain + Pydantic: Built a resilient LangChain-based Q&A application with Pydantic-validated structured responses, graceful error handling, and parallel batch processing for reliable and efficient query processing.

Tech Stack: Python, LangChain, Pydantic, LLMs

5.Weather Assistant with MCP | Python + FastMCP: Built a custom MCP server using Python and FastMCP that connects AI assistants to the National Weather Service API for real-time weather information. Integrated and tested MCP tools with Claude Desktop and Cursor.

Tech Stack: Python, FastMCP, MCP, NWS API, Claude Desktop, Cursor

6. Data Analysis & Transformation Pipeline | Python & Pandas: Built a Python-based data analysis pipeline using Pandas and Jupyter Notebook, covering data cleaning, transformation, filtering, sorting, aggregation, ranking, time-series resampling, and multi-format data export.

Tech Stack: Python, Pandas, Jupyter Notebook

EDUCATION:

BTech - Godavari Global University, JNTU(Kakinada) - 2003-2007 | GPA: 64%

Intermediate - Siddhartha college - 2001-2003 | GPA: 80%

INTERESTS:

Reading Books, Listening to music